

Git Local Commands and Workflows

Setup:

1. git clone <https://github.com/yanivkrol/git-training.git>
2. git config (--global) core.editor "/Applications/IntelliJ\ IDEA.app/Contents/MacOS/idea --wait"
3. JetBrains commit window setup

Good to know

- `4fh6k3df` - Example of commit hash
- `HEAD` - Reference to the latest commit in the working branch
- `master` / `main` - Reference to the latests commit in master/main
- `<hash>^` - Reference to one commit prior to the commit with the given hash. E.g. `4fh6k3df^`, `HEAD^`. Adding more `^` will go further backwards, for example, 3 commits behind is `<hash>^^^`
- x number of `^` s can be replaced by `~x` . e.g. `HEAD^^^` == `HEAD~3`
- A branch name is also a reference to specific hash (the HEAD of the branch)

Tips and tricks:

- `HEAD^` or `HEAD^^` can be used to quickly reference one of the latest commits in the working branch.

Commands

git fetch

Fetch changes in the repository without merging them.
This includes new branches, commits, and tags.

`git pull`

Performs `git fetch` and then tries to also merges/rebase (configurable) onto the current branch.

By default, uses `origin/current-branch` as the remote branch to pull from.

Options:

- `<hash>` specify a different branch for pull (and also try to merge, careful!)

`git push`

Push changes of the current branch to remote

Options:

- `<hash>` specify a different branch to push
- `-f` Force push when history changes (amend or rebase)

`git checkout <hash>`

Changes the working branch to the specified commit. Usually used with branch name, but can be used with a hash as well.

Options:

- `-b <name>` - Creates a new branch with the specified name

Tips and tricks:

- like the `cd` command, `git checkout -` checks out the previous branch

```
git status           > branch1
git checkout branch2 > branch2
git checkout -        > branch1
```

git status

Displays the state of the working directory and the staging area. Also shows statuses of merges, rebases, cherry-picks, etc.

git log

Displays history of the repository. Can be used to quickly see how our branch looks and look for specific commits to use `fixup` with.

Options:

- `-x` - Limits the number of commits shows, counting from HEAD inclusive (`git log -5` - shows last 5 commits)
- `--oneline` - One line per commit. Displays only the commit hash, branch name, and commit message.

Tips and tricks:

- `git log --oneline master..HEAD` - to fully show current branch

git branch

Shows branches (local only by default)

Options:

- `-m <new-name>` - Rename branch
- `-d <branches...>` - Deletes merged branches (merged into current branch)
- `-D <branches...>` - Like `-d` but "forced", deletes unmerged branches too

```
git add <files> <directories>
```

Stages specified files nad directories.

Tips and tricks:

- Staged changes using the JetBrains commit window will be actually staged (as opposed to the regular commit window). This lets us easily stage **specific lines**

git commit

Captures a snapshot of the project's currently staged changes.

Options:

- `-m` - Specify a message (`git commit -m "my message"`)
- `-a` - Commit all modified files (automatic staging)
- `--amend` - Adds the changes to the last commit, also editing the commit message. If used without changes, only edits the message. `--no-edit` Can be added.
After an amend, the commit hash changes, which means you'll have to `push -f`
(Assuming original branch is on remote)
- `--squash <hash>` - Creates a commit that will later be "squashed" together with the commit specified, merging their changes and their messages
- `--fixup <hash>` - Same as `squash` but using the specified commit's message instead of merging them. This will be in greater use than `squash`

git commit

Tips and tricks:

- `git commit -a --amend --no-edit` - quickly add a missing semi colon, or any other changes for which there's no reason to create a new commit
- `git commit -a --amend --no-edit && git push -f` - Same as above but when you've already pushed (for example build failed because of missing colon)

`git reset <hash>`

Change the HEAD of the branch to be the specified commit, all "lost" changes will be unstaged

Options:

- `--soft` - All "lost" changes remain staged
- `--hard` - All "lost" changes are **permanently removed** (be careful)

git stash and git stash pop

Temporarily shelves (or stashes) changes you've made to your working copy so you can work on something else, and then come back and re-apply them later on

Options:

- `-u` - Also stash new files

Tips and tricks:

- Don't use unless you are going to pop in the next 2 minutes (It's a black hole).

Alternative:

```
git add . && git commit -m "stash"  
do stuff...  
git reset HEAD^
```

- Can be used across branches

`git rebase <hash>`

Moves current branch to be based of the specified commit. "Rewriting history"

Options:

- `-i` `--interactive` - Enables controlling what happens with each commit above the base commit
- `--autosquash` - When using `-i`, automatically moves squash or fixup commits to their correct places in the interactive editor

`git rebase <hash>`

Tips and tricks:

- `git rebase master` - To stay updated
- `git rebase master -i (--autosquash)` - Interactively edit the entire branch
- After a rebase, the commit hashes change, which means you'll have to `push -f` (Assuming original branch is on remote)
- When rebasing a remote branch and the history was changed, use `-i` and drop all local "duplicate" commits (otherwise conflicts will be horrible)
- If at any stage during a rebase you are not 100% sure how to resolve a conflict or got lost, `git rebase --abort`

`git cherry-pick <hash>`

Adds the specified commit to the top of the current branch

Tips and tricks:

- Can be used to apply temporary changes that we store in dedicated branches

```
git checkout -b stg-settings
git commit -m "stg kafka settings"
git commit -m "stg db password"
```

```
...
```

```
git checkout feature-branch
git cherry-pick stg-settings
```

```
...
```

```
git reset <HEAD original hash>
```

> Assume a few commits already

> We now have the staging settings in our branch

> discards the staging settings